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Market Dominance as a Precursor of a Firm's Failure: Emerging Technologies and the Competitive Advantage of New Entrants

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ABSTRACT: New entrants in many industries are able to challenge the business of historically dominant firms. In many markets, dominant players have pursued pricing and service policies that, although once highly effective, now make their markets attractive targets for aggressive new entrants. The entrants' strategies rely on lower overhead costs, new technologies, alternative distribution channels, and the active targeting of profitable customers. Several factors will make it possible for entrants to attack dominant players; simplistic historical pricing mistakes or policies of promising or providing universal service will make it attractive for new entrants to attack. Restrictions on the flexibility of incumbents—both externally and internally imposed—may make it difficult for dominant players to defend themselves effectively against attack by more flexible entrants with cream-skimming strategies and newer technology. We develop a set of alternatives for incumbent firms facing increasing “contestability” in their markets and the threat of agile entrants.

KEY WORDS AND PHRASES: banking, contestability, death spiral, dominant firms, insurance, market entry, market segmentation, strategic use of information.

INFORMATION TECHNOLOGY (IT) IS A GROWING FACTOR IN PRODUCING and distributing goods and services. Through outsourcing, service bureau contracting, and new,

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“virtual” organizational forms, IT has reduced the costs of entry into many industries. The result is that even dominant firms are pressured to keep prices low to deter entry, and to be prepared to employ credible (i.e., fear-provoking) price reactions to the challenges of new entrants. IT also makes it possible to segment customers accurately, to distinguish those that make significant contributions to profitability from those from whom revenues cannot cover expenses. New entrants in many industries readily apply IT and data-based marketing techniques to attract the most profitable market segments. After a new entrant’s attack, the established, historically dominant provider finds its customer base eroded and its profitability correspondingly reduced. This paper describes the problems of increasing “contestability” and targeted cream-skimming entrants and explores the responses available to established firms.

We have argued previously [4] that dominant players frequently rely upon a practice of uniform average pricing, even when the costs to serve individual members of the customer base differ greatly. We have shown that this simplistic pricing strategy attracts new entrants, who often employ more sophisticated pricing strategies. Dominant players who continue to pursue average-cost pricing rather than adopting segment-based strategies face growing threats. The loss of revenue to these new entrants, coupled with a naive response from the dominant firms, creates a self-reinforcing loss of additional market share—a phenomenon we call “death spiral.”

Examples of Targeted Entry

IN A WIDE RANGE OF INDUSTRIES, DOMINANT PLAYERS ENJOYING near-monopoly positions (but required—whether by regulation, corporate culture, or myopia—to provide universal access to their services) can be seriously damaged by cream-skimming by new entrants—that is, their efforts to serve only customers in the most attractive segments of the market [4]. An insurance carrier such as USAA, for example, is able to cream-skim through its access to a preferred risk pool: namely, military personnel and their families. Because of known economic, behavioral, and lifestyle tendencies, this group files fewer claims and can profitably be offered lower rates.

The fact that Blue Cross allows its customers to continue to use their current physicians, while many HMOs do not, creates an advantage for HMOs: Patients whose medical needs require frequent visits to health-care providers are the patients who are more likely to have developed close relationships with their current doctors, and thus especially value the ability to stay with their physician when choosing a health plan. Blue Cross, thus, will systematically retain patients who require regular medical care, while HMOs will attract proportionally more of the patients who require fewer services and are thus less expensive to serve. HMOs further attract their targeted customers through more comprehensive “wellness” coverage. To break even in the face of this unpleasant selection bias among its customers, Blue Cross must raise premiums, increase its deductibles, or reduce coverage—all of which accelerate the loss of attractive customer segments to HMOs.

Moreover, emerging IT applications will enable HMOs to project costs and to target their selected patients even more accurately, placing even greater pressure on existing

insurance carriers. If these existing carriers naively increase rates further in an effort to preserve their profitability, they will accelerate their loss of market share in the attractive segments while creating additional opportunities for competitors to cream-skim, placing still more pressure on their margins.

Average-cost pricing can provide comfortable profitability when customers are representative of the entire market and when competitors are not cream-skimming. When new entrants can capture market share by offering preferential pricing to the most attractive customers, established players compound the damage with naive responses. Ultimately, the use of average-cost pricing on the remaining, more costly-to-serve, customers requires the established player to raise prices, driving away more accounts. Without an aggressive response to the cream-skimming new entrants, the established player self-destructs.

Examples of Cream-Skimming

The threat of “death spiral” is particularly evident in mature industries, such as credit cards and insurance, whose customers differ significantly in their profitability to the service provider.

Credit Cards

Credit card customers with reliable incomes who borrow actively on the card have above-average profitability. Tenured schoolteachers, for example, have long-term secure incomes and a correspondingly low risk of default; these teachers predictably earn less in the summer months, leading to temporary borrowing to be repaid during the academic year. Using predictive demographic data, card issuers such as Signet Bank have targeted low-risk, high-income borrowers, enticing them away from established bank card issuers by offering lower annual fees and reduced interest rates. The established players are left with fewer profitable accounts, and, unless they respond aggressively, will find themselves serving only the least attractive segments.

Insurance

Insurance risks and claims experience vary across customers and customer segments [11]. DirectLine, a subsidiary of Royal Bank of Scotland, has identified several lower-risk segments of the U.K. property and casualty insurance market. By avoiding segments of the market likely to generate claims—such as male, unmarried automobile drivers under the age of twenty-eight—lower prices can be offered to the rest of the pool. DirectLine chooses its prospective customers using demographic data and solicits their business directly through mailed marketing materials that highlight substantial reductions in premiums for customers switching to DirectLine.

We have been surprised at how widespread the cream-skimming/death spiral phenomenon appears to be, and even more surprised at how rarely dominant players seem to be able to develop effective responses. This paper details the prerequisite

factors for death spiral, the role of IT in enabling or facilitating each of them, and the conditions under which it may be impossible for the incumbent to respond.

Case Example of Contestability and Pricing Cross-Subsidy: USWest

THE ARENA OF COMPETITION FOR TELECOMMUNICATIONS SERVICES in both urban and rural regions highlights the vulnerability of established firms facing cream-skimming entrants. Many cities in otherwise relatively less-populated regions—such as Denver, Colorado—are attracting aggressive new providers of telephone and telecommunications services who expect to compete on favorable terms with the incumbent provider. (Lower Manhattan has *five* competitors offering local phone service as of this writing.) Denver's population density and commercial activities make that city's customers cheaper to serve than those elsewhere in Colorado. Given that uniform pricing is imposed by state regulators, the Denver customers are, unsurprisingly, the source of most of USWest's profits in the state. The new providers plan to undercut USWest's rates in Denver, thus gaining market share.

USWest faces the dilemma of either matching their new rivals' reduced rates or losing market share in their most profitable segment. Historically, USWest has been forced to use the profits from Denver to subsidize the rest of Colorado's higher-cost customers. Ironically, this artificial "price umbrella" in Denver allows attackers using new technologies to have the low *price* offerings in Denver even when USWest is the low-*cost* provider with its large fixed investment in infrastructure. The options open to USWest are:

- Cut prices in Denver while leaving the rate structure for the rest of the state unchanged, undermining revenues and profitability. Regulators, however, could pressure USWest to cut their rates statewide for fairness.
- Lose share in Denver and apply for rate increases elsewhere in the state—a naive response that opens the window for death spiral by making the second-most profitable city (e.g., Boulder) a target for entry. Furthermore, regulators may choose not to approve the proposed increases.

It appears that USWest's existing market share, sunk investment in wire and switches, and regulatory obligations to serve its various customers "fairly" (under its regulators' concept of fairness) present a significant impediment to its stable, future profitability. In the long term, new entrants will attack USWest's most attractive markets, using new technologies such as asynchronous transfer mode (ATM) and switched video. Whether these new technologies could provide service to 100 percent of Colorado customers at low cost is irrelevant; unlike the twisted-pair copper wire and hierarchical switching structure owned by USWest, the new technologies are easily scaleable and open new routes and toll structures, allowing a new entrant to continue to nibble at USWest's most profitable customer segments until the previously dominant firm either makes a radical shift in strategy or collapses.

Critical Factors and Strategic Responses

THE PROTOTYPICAL PROBLEM FACING DOMINANT, ESTABLISHED service providers with heterogeneous customers has three components:

- *Newly contestable market*: Alternative providers are able to offer similar or superior services to the dominant firm's current customers, who face low or nonexistent switching costs. The new providers can enter and exit the market at minimal cost because of low fixed-cost facilities contracting or use of a virtual organization design [9, 10].
- *Historical average-cost pricing despite differential cost structure*: Established players once could set prices based on the average cost of serving a broad range of customers. This implied cross-subsidy (e.g., Denver phone customers' rates supporting below-cost service to remote areas) has attracted new entrants to precisely those segments (and solely those segments) now paying prices greater than those justified by costs.
- *Exit barriers and flexibility restrictions on dominant players*: Regulators concerned with public welfare will discourage attempts by established players to withdraw from unprofitable regions or eliminate service to costly-to-serve market segments.

Contestability

We find the prevalence of the economic conditions of *contestability*—the ability of new entrants to easily slip in and out of markets without sinking massive capital investments or committing irrevocably to future market presence or strategies [10]—is intensified by the winding down of regulatory monopolies, by the emergence of new, low-cost distribution channels, by the introduction of new technologies that reduce the minimum required scale to compete in an industry, and by placing or enforcing restrictions on incumbents that restrict repricing or prevent exit from certain segments of the market. Contestability can arise, for example, when the monopoly on scale advantages enjoyed by larger firms is eliminated through the creation of a new outsourcing market for these large-scale technologies. For instance, scaleable ATM, cable, and wireless technologies make it significantly easier to enter a market as a targeted telecommunications company. Insurance policies for Royal Bank of Scotland's DirectLine are marketed directly to selected households without an agent or other intermediary. Such cost advantages in distribution, over and above the underwriting advantage generated from accurate customer data, enable DirectLine to further undercut the premiums charged by established players using traditional distribution channels. Exit restrictions are often imposed on established firms by regulators and public officials eager to maintain "fairly" priced telephone, banking, and other services in all areas and to all customer segments. Dominant firms may not be able to cut back services selectively to unprofitable segments or regions; even if they become uncomfortably aware that

dominant presence for these segments is undesirable, their retreat may be cut off by regulatory restriction.

Our treatment of contestability differs from previous treatments. Contestability theory in economics [1, 2, 9, 10] has traditionally been concerned less with strategic recommendations for threatened incumbents than with determining public policy, regulatory action, and antitrust recourse against currently dominant firms. Contestability models, furthermore, ignore cream-skimming, assuming instead that entrants will follow the same single-price strategy as incumbents—differing only in that the single price offered would be lower. Fudenberg and Tirole [7], in evaluating older theory, point out that contestability theory involves ascribing “superpowers” to entrants: the ability to enter a market, steal customers from incumbents, and retreat before the incumbent introduces a new strategy. There are limits to contestability and not all entrants possess such superpowers, but we submit that certain entrants do have the ability to capture market share quickly from incumbents: These abilities result from the coincident combination of historical mispricing, sluggish responses by incumbents, IT’s ability to support segment-specific strategies, and the rise of an outsourcing industry that spreads large fixed costs (of data processing, distribution, capital expansion, and expertise generation) over multiple firms, enabling many entrants to execute such a profitable “hit and run” strategy without committing to universal service.

Differential Cost and Heterogeneous Customers

The problems facing established players are exacerbated by differential costs for serving market segments combined with average-cost pricing policies that effectively overcharge attractive customers to cross-subsidize other customer segments. We now bluntly refer to average-cost pricing as “simplistic historical pricing errors.”

Differential cost structures are generally compatible with average-cost pricing only in the presence of an absolute entry barrier—for example, regulatory policy specifically preventing entry or a prohibitively high cost associated with accurately discriminating among customers based on information. Stock markets such as the New York Stock Exchange (NYSE) are regulated by the Securities Exchange Commission (SEC) and provide a form of universal service, executing buy and sell orders from all traders [6]. Traditionally, Rule 390 has required Exchange member firms to bring their customers’ orders to the floor of an exchange. In recent years, off-exchange “third markets” run by firms that are not members of the NYSE have attracted order flow by cutting and even rebating commissions on low-risk trades. Third-market dealers “profile” their order flow to comprise only low-risk orders. By choosing the customers from whom they will accept orders, third-market dealers threaten the average-cost pricing approach used by the NYSE. The combination of average-cost pricing and nondiscriminatory order handling (i.e., not attempting to screen arriving orders by riskiness) at the NYSE becomes untenable as the Exchange loses more small, low-risk orders, and as the costs associated with serving the remaining order flow increase. Raising rates to reflect higher costs will, of course, only drive more orders off-exchange, continuing the death spiral.

Insurance providers face differential costs in serving prospective policy holders [11]. Age, gender, and smoking habits are often among the criteria used to set premiums. Life insurance policies, for all their complexity, are still priced without regard to certain other data that can further refine the classification of an individual's level of risk. For instance, the life expectancy of right-handed people is slightly higher than that of lefties. No insurance carrier known to us offers preferential rates to righties, but if one did so successfully, the remaining carriers would find a disproportionate number of the higher-risk, less profitable left-handed customers in their pools.

Established firms with heterogeneous customer bases have historically spent little time determining which customers are most expensive to serve, and/or which are the best revenue producers. This inattention creates an opportunity for attackers to develop predictive statistical models to direct targeted attacks on the most profitable segments of the market; these attackers' models, while certainly less accurate than the best models that established players could theoretically develop to exploit their existing data advantage, are frequently far superior to the targeting mechanisms employed in practice by dominant firms [13]. Through continued inattention to modest segmentation opportunities and average-cost pricing, dominant players create conditions that encourage attack. When combined with ease of entry, dominant players without effective responses face death spiral [4]. We explore below why defenders may fail to develop effective responses.

Flexibility Restrictions

Traditionally dominant firms frequently face restrictions—whether regulatory or self-imposed—on pricing and/or on market exit that, in the presence of new, cream-skimming entrants, make it difficult to retain profitability. Certain *regulatory policies* (including restrictions on branch closings for banks and mandated universal service for local telephone providers) make it difficult to scale back unprofitable operations in regions where a dominant firm faces a competitive threat. Legal contracts that cannot be canceled or terminated without cause (e.g., “whole life” insurance policies, which cannot be repriced for existing policy holders) limit a firm's ability to retrench or reprice to eliminate cross-subsidization from its strategy.

Corporate governance or culture may restrict organizations in responding to the threat of death spiral (e.g., the NYSE's historical role has been to accept all customer orders without distinction or price discrimination). Traditional methods of distributing products, especially those that are human-resource-intensive, will also limit changes in strategy (e.g., the use of an agency sales force to sell insurance cannot easily be terminated without major disruption in current sales as well as considerable explicit restructuring costs).

Moreover, in the absence of strategic flexibility, *sunk costs* such as in-place switches and local-loop telephony equipment can become “stranded assets” with little value. Established firms with fixed technologies facing more flexible entrants with new technologies (or a wide choice of available traditional technologies, with no commitments made yet) have little chance to compete effectively; given that these incumbents

have been targeted by the entrants in the first place, their technology is quite likely either obsolete or vulnerable to cream-skimming.

Finally, the *information infrastructure* itself (or lack thereof) may represent a significant restriction on the flexibility of established players. Accounting systems developed in the early stages of data processing will not provide the information needed for customer profitability analysis; such legacy systems qualify as stranded assets even though they may be fully depreciated.

Strategic Options and the Role of Information Technology

Uniform, average-cost pricing used by the established players is sufficient to provide comfortable profitability when the established players' customers are a representative sample of the broader market—that is, when no competitors are deliberately or inadvertently cream-skimming. Entrants, on the other hand, have strong incentives to capture market share by offering preferential pricing to only the most attractive customers. The factors noted—newly contestable markets, differential cost structure, and flexibility restrictions—combine to make it difficult for a dominant firm to respond to cream-skimming new entrants. However, several strategic options are available to established but threatened firms. Active responses that we expect to be effective are to *reprice*—eliminating cross-subsidies—and to *regain flexibility* by introducing technology-enabled low-cost distribution channels—such as self-service banking kiosks or toll-free service lines to reduce the need for sales agents and field service offices—to restructure the expense of serving customers. The firm's choices of strategies include:

- *Death spiral* (the default option): A dominant firm that does not respond actively will fall into death spiral. Profitable customers will be lost, and market share in attractive regions will shrink.
- *Break away from historical pricing mistakes*: Repricing products and services to reflect their costs (or to reflect the value created for the customers who consume them) reflects an active response, ending the cross-subsidy that created the cream-skimming opportunity in the first place. Unprofitable customers will have to pay higher fees and face increased rates, while charges to attractive regions and profitable customers will be reduced in order to protect this market share from poaching by entrants.
- *Regain flexibility*: A threatened firm can adopt another active response: Adjusting its cost structure by adopting new technologies or opening up new, lower-cost distribution channels. The dominant but threatened firm may need to shed infrastructure of stranded assets—including bank branches, insurance sales agents, and historically expensive but outdated or inflexible technologies—to establish a cost structure that can compete favorably with that of the entrants.

In implementing these options, IT can play a number of roles. For instance, when used to eliminate the information asymmetry between a securities trader and his or her exchange counterparties, IT has been used both to correct historical pricing

mistakes and to gain flexibility. In securities trading, a trader's impatience to execute an order is often a good indicator of the riskiness and information content of his or her trade. Two of today's most successful off-exchange trading systems, POSIT and the Crossing Network, are based on batched "crossings" of submitted buy and sell orders. Since the crossings occur only at a few selected times each day, a trader using such a system is effectively and credibly "signaling" that it has no information that is critically time-sensitive, and thus it is an attractively low-risk trading partner [6, 13]. Thus, a trading system can be used to provide the opportunity for traders to signal; the availability of this opportunity reduces the information asymmetry faced by other traders. Other traders now are encouraged to price a patient (safe) order *more favorably* than the price they charge a higher-risk order that is slated for immediate execution.

Examination of Additional Case Studies

Hong Kong Retail Banking

THE RIVALRY BETWEEN THE HONG KONG AND SHANGHAI BANK CORPORATION (HKSB) and Citibank in Hong Kong is a prime example of a cream-skimming entrant successfully attacking an established player. Notably, Citibank has not attempted to overpower HKSB by targeting HKSB's entire customer base. Rather, it has developed distribution channels for products and services that are attractive only to certain customer segments—unsurprisingly, this group contains almost all customers whose service is profitable for HKSB. The newly contestable consumer banking market in Hong Kong has opened up an avenue for cream-skimming by Citibank, which is now HKSB's most dangerous and profitable rival.

HKSB's customer base is characterized by "Love 'em's" and "Kill you's." It is believed that 20 percent of its customers generate more than 100 percent of HKSB's gross profits and that 60 percent of the customers actually cost HKSB money. A typical "kill you"—a customer who reduces firm profitability—is a low-balance retail customer who performs his or her transactions in the branch, in person, using a human teller. HKSB faces several of the critical factors identified above that will erode its profitability and make it difficult to formulate a response:

- Exit restrictions prevent HKSB from easily restructuring itself to defend its profitability. Used brick-and-mortar bank branches are not highly valued at present. HKSB's infrastructure, especially its branch network, cannot be easily liquidated for its book value, even though the value of the branches will diminish as Citibank takes away more customer accounts.
- Public pressure commits HKSB to maintaining an extensive branch network and to providing low-cost banking services to the broad population in Hong Kong.

Strategic options open to HKSB are:

- Accept death spiral: In the absence of an effective response, the profitability of HKSB's consumer bank will fall as rivals target the attractive customer seg-

ments and leave HKSB the costly-to-serve, low-revenue “kill you’s.”

- Adopt differential pricing strategies such as charging small accounts a fee for every in-person access to branch tellers and applying fees to low-balance accounts to recover the costs of providing services. Established players frequently object to such strategies because they drive away customers—but driving away a current money-losing customer is as profitable as recruiting a new money-making one.
- Pursue technological alternatives, including new, alternative means of distribution and cost-reducing alternatives that do not exclude the “kill you’s” entirely. HKSB’s ability to respond will depend on its ability to distribute banking products with fewer tellers and branches and on encouraging use of automated teller machines, kiosks, and telephone service centers, especially for small-balance accounts.

Citibank’s attack on HKSB is an example of what, in U.S. telecommunications, we call the “Donner Pass” scenario: Citibank, under pressure as a dominant incumbent in the United States, attacks as a new entrant outside its region. Similarly, the regional Bell operating companies (RBOCs) in the United States are invading one another’s most attractive market segments—cellular, cable, network facilities management, and high-capacity teleconferencing services.

Off-Exchange Trading Systems in London

Until recently, the London Stock Exchange (LSE) faced few threats to its control of nearly all equities trading in the United Kingdom. A new rival, TradePoint, which began operations on September 21, 1995, is attempting to attract trading to its fully electronic trading system. TradePoint is another example of a new entrant seeking to cream-skim attractive business from a dominant organization, and its impact on the viability of LSE member firms could be significant [6].

In the United States, off-exchange systems have attracted investors who use passive, inexpensive trading methods such as indexing. In a competing dealer market such as the LSE, investors buy from and sell to dealer intermediaries (the LSE’s market-maker firms) at quoted bid and ask prices. Historically, financial intermediaries in London took the other side of all trades; their profits from dealing with customers whose trades posed little risk to them more than covered their occasional losses to customers whose superior information made their trades more risky (and less profitable). The LSE faces competition from TradePoint and several other recently introduced “fourth-market” off-exchange systems that match investors’ orders, *without* the involvement of a broker or a dealer, at a price halfway between the bid and the ask from the LSE. Developing an effective response is difficult for the LSE’s member-firm intermediaries because investors frequently have a better understanding of the reasons behind their trades and hence of the riskiness of their trades for the market makers, than do the market makers who deal with them.

Today, with alternatives designed primarily for less risky (uninformed) trades

available, the profitable customers of LSE member firms are likely to take their trades increasingly elsewhere, leaving market makers with a higher proportion of loss-making business. Without providing member-firm intermediaries with a trading mechanism for distinguishing safe from risky trades, naively granting low-cost execution to protect market share (even among attractive customers) leads to catastrophic losses [6, 11]. Strategic options available to the LSE and its membership are:

- **Accept death spiral:** The Exchange members face growing losses, and Trade-Point captures significant low-risk trading volumes. Spreads widen and more investors migrate to fourth-market systems, aggravating the market-makers' problem.
- **Adopt price flexibility:** By repricing their services, LSE market makers can attempt to retain order flow from safe investors. Unfortunately, a dealer will not be able to infer the true motivation for an investor order, and hence will not know its level of risk, until it is too late. Simplistic repricing strategies are not possible if LSE market makers are required to serve all traders, or if they cannot accurately discriminate among the informed and the low-risk traders. Only if investors can credibly signal the low risk that would be incurred by the counterparty to their orders can the dealers price accurately and the LSE maintain its position as the central market for equities trading in London.
- **Pursue signaling systems for LSE order handling and routing:** Simple signaling mechanisms that reduce or eliminate information asymmetries do exist [12]. Clemons and Weber [6] show that systems that allow market makers to infer a signal and perform differential pricing for their intermediation services can avert market makers' losses and preserve the centralized market's dominant position—provided these systems can be implemented despite cultural and regulatory barriers.

While IT creates an off-exchange trading threat by easing the way for entrants, systems also enable sophisticated pricing strategies for the LSE dealers, based on credible signals received from investors. Note that the LSE could avert death spiral and maintain a single central market by both embracing technological alternatives and adopting the strategy of the off-exchange providers by providing differential pricing for different types of order executions. Omitting either will lead to continued fragmentation and eventual collapse.

Signet Bank: Credit Card Issuance

Bank card issuers also have “Love ’em’s” and “Kill you’s.” An issuing bank’s profits depend on the proportions of its cardholders who pay their balance in full each month (“transactors”), borrow at the rate set by the issuer but eventually repay (“revolvers”), or default on outstanding balances (“defaulters”).

Recognizing the variation in profit across customers, Signet Bank uses a cream-skimming strategy to get Signet cards into the hands of carefully targeted customers—usually high-balance “revolvers” with reliable incomes. Signet, now a major issuer of

MasterCard and Visa cards, uses tailored promotions to win such “love ’ems” away from current card issuers, including major card-issuing banks such as Citibank and Chase. Firms use initial promotions designed to overcome switching costs, offering cards with low APRs and reduced (or eliminated) annual fees. Signet’s mailings sometimes assure customers that they will be allowed to skip monthly payments at some points during the year if they so choose; finance charges, of course, would continue to accrue, making customers such as the aforementioned tenured teacher especially attractive.

The strategic options available to dominant card issuers are:

- **Accept death spiral:** Competitors unable to identify their profitable accounts, or unwilling to offer them preferred rates once identified, are increasingly losing these customers; ultimately, dominant issuers may find themselves forced to raise their annual fees or interest rates on their remaining customer base to maintain profitability, accelerating the loss of profitable customers and collapse of profitability.
- **Pursue differential pricing:** This strategy may be easier to implement in card issuance than in, for example, insurance, because a card issuer has more freedom to dynamically reprice its portfolio after initial card issuance. Much of the industry seems to have adopted a preliminary form of differential pricing for the best customers through fee and rate competition but, without a sufficiently widespread change in price or fee structure for the remaining accounts, profitability rates have eroded. Rather than restructure their own offerings, suffering card issuers are tempted to pursue competitors’ customers; the “Donner Pass” scenario unfolds.

Strategies Available to Previously Dominant Firms

WE EXPECT THAT MORE CONTESTABLE MARKETS WILL EMERGE, just as they did in the examples discussed here. The problem of cream-skimming entrants attacking dominant firms is general, visible in a range of industries, and surprisingly difficult to address. Firms that, due to their dominant positions in their markets, appear secure will become increasingly vulnerable to as-yet-unforeseen potential entrants, if either (a) the dominant firms lack accurate cost and revenue data to know which customers and market segments are profitable [13], or (b) regulators, social policies, or existing contracts require them to continue average-cost pricing in the presence of differential costs. We also expect that dominant organizations in a range of industries will struggle to respond to new conditions of contestability for the following reasons:

- The public’s expectations for universal service as an entitlement will make it difficult for previously successful companies in critical industries such as banking, insurance, and telecommunications either to reprice certain segments of the market or to abandon them entirely. The very success of the incumbents, and their acquisition of broad coverage in all segments of the market, may thus be transformed from a strength to a weakness. In the spirit of “judo economics”

[8], we expect firms to be stripped of the ability to execute a strategic retreat and forced to defend untenable positions at extraordinary cost.

- The incumbents' existing "mispriced" contracts with customers similarly reduce their future pricing flexibility without affecting the pricing strategies available to new entrants. Incumbents have effectively contracted with customers, granting the option to the customer, at his or her discretion, to continue or terminate the relationship with the incumbent—contracts that cannot, unfortunately, be canceled at a moment's notice when the market turns. Again, success in obtaining customers may have become a source of unfortunate obligations that cannot easily be canceled.
- The incumbent's corporate culture and the control exerted by an entrenched sales channel (for instance, a commission-earning agent force) that holds hostage the company's revenue generation in the short term, will limit established companies' distribution options and flexibility. Again, success may have created a culture that relies upon strengths that are difficult to abandon, even though they now hinder rather than support future success.
- The incumbent's reliance on outmoded infrastructure—such as expensive brick-and-mortar bank branches and other costly forms of distribution—and their desire to avoid "stranding assets" at any cost will further limit flexibility, both in conceiving future strategies and in executing them. That is, the mere desire to avoid writing off investments in branch networks or other outdated assets (even though these assets might otherwise be replaced with useful ones at surprisingly low cost) may reduce firms' pricing options and exit flexibility, inasmuch as managers refuse to entertain the idea of these massive writedowns. Once again, the dominant firms' success in acquiring market share and the fixed assets to support it may have been transformed from a source of strength to a long-term weakness.

In brief, several factors make it *possible* for entrants to attack dominant players; new technologies for distribution and contracted services in a "virtual" organization, simplistic historical pricing mistakes, or policies of promising or providing universal service make it *attractive* for new entrants to attack, especially in the most profitable market segments. Restrictions on the flexibility of incumbents—both externally and internally imposed—may make it *difficult* for dominant players to defend themselves effectively when attacked by more flexible entrants with cream-skimming strategies and newer technology. The following are suggestions that an incumbent might pursue to develop an effective strategic response:

- *Take preemptive incremental actions*, such as reducing costs to efficient levels, *before* the threat arises, with an eye toward improving the ability to withstand the attack of a future cream-skimming entrant. Recent mega-mergers in banking, such as the merger of Chemical Bank and Chase Manhattan, are aimed at achieving increased efficiency through the consolidation of branch networks and the closing of redundant facilities; thus, these mergers may be seen as part of a cost-control strategy that will both improve short-term results and purchase

additional time to respond to a future competitive threat.

Such incremental improvement will not suffice, however, to deter or to compete with potential entrants who possess significant structural advantages, such as more flexible cost structures due to alternative distribution and/or more efficient revenue generation due to more precise customer segmentation and more accurate service pricing. Dramatically different cost-cutting strategies for surviving impending shakeouts are arising in telecommunications; the U.S. telecommunications companies with the widest distributions of costs of providing local service have actually considered shedding their highest-cost regions—whether by selling them, giving them away, or paying others to take them—to escape the burden imposed by regulators' uniform pricing policies.

- *Develop alternative distribution systems*, including branchless home banking, wireless local telephone service, agent-less distribution of insurance products, or a new intermediary-less screen-based financial trading mechanism. This development process entails a fundamental reexamination of products and services to be offered to different market segments and a segment-by-segment analysis of which means of distribution will be most effective (and/or least expensive) to serve them—a radical departure from incremental cost-cutting strategies classically employed to survive industry shakeout. The chief executive of at least one major U.S. insurance company has concluded that new entrants with lower-cost alternative distribution systems will eventually force his company out of the market. He has decided, accordingly, to develop a direct distribution arm, with the idea of increasing the chances that the attacker that eventually destroys him will at least be owned by him.
- *Increase pricing flexibility*, to allow emulation of the cream-skimming strategy of potential entrants. Such flexibility requires far better quantitative information on customers than is available from the knowledge base of most service organizations, performing far more detailed market research, and developing databases and applications specifically aimed at supporting the development of new products and services, along with their associated pricing strategies. Entrants will be deterred if they expect a rapid pricing reaction by the incumbent as a result of their entry. Pricing flexibility and rationality are becoming vitally important to banks, securities firms, and even stock exchanges as they prepare for increasing competition from electronic off-exchange dealing systems.
- *Work more effectively with the news media*, especially to present the case for differential pricing to the press as the only option to exit. Incumbent banks that impose fees and service charges on their unprofitable customers, or local telecommunications providers that cut rates in big cities and increase them in remote rural areas, usually receive negative publicity; such moves could equally well be seen as defensive actions aimed at protecting the firm, its customers, and its shareholders from the attack of opportunistic new entrants that are “freeloading” in the market.
- *Learn to understand the threats*, and to manage the organizational resistance and risks associated with strategic change. Large successful firms should seek to

gain extra time, either as unchallenged incumbents before entry begins or as bloodied but unbowed competitors, to decide which segments of the market are both profitable and defensible, and should prepare for increased contestability by using their information advantage as incumbents to identify, occupy, and lock up these desirable niches.

Research Plans

OUR ONGOING RESEARCH IS AIMED AT MODELING THE COMPETITION between incumbents and new entrants, and considering the factors that would enable us to predict success of alternative strategies. We model the following general conditions, using parameter values and stylized technological improvements inspired by specific industries:

- *Ease of attack*: How has new distribution technology facilitated new entrants' abilities to begin to attack existing markets? What effects have changing economies of scale or increased outsourcing opportunities had on ease of entry? These factors combine to determine the speed with which new entrants can gain the capability to serve the market.
- *Attractiveness of attack*: How have pricing strategies of incumbents created opportunities for cream-skimming by new entrants? What cross-subsidies exist, and how can they be rationalized or reduced? How large are these opportunities? How significant will the differences between entrants' and incumbents' services appear to customers? How much improvement over the status quo is required to encourage customers to shift their business to new entrants? These factors combine to give us the speed with which as-yet-unarrived entrants could theoretically capture market share under existing conditions.
- *Difficulty in defense*: How effectively can regulated incumbents gain relief, letting them change their universal service or uniform pricing strategies? How can they shed underperforming assets, outdated distribution networks, or inappropriate labor-intensive practices? In short, how can they come to resemble new entrants, to the extent they wish to? These factors combine to give us the speed with which incumbents can reposition themselves to withstand attack.

To examine the expected outcome of alternative strategies and the opportunities for entry deterrence or accommodation under these varied conditions, we are developing a strategy-based industrial dynamics model. In our ongoing research, we have classified the generic strategies available to incumbents as follows:

- *Harvest and Exit*: Make money while you can and exit gracefully, ideally by selling for an attractive price to someone who does not yet realize the difficulties you face. The recent "sales" of PacTel and NYNEX appear to reflect this strategy.
- *Copycat*: Endure the pain and the significant financial losses over the short term; shed underperforming assets, unprofitable customers, and inappropriate com-

mitments; and learn to duplicate the actions of the new entrants that are tormenting you.

- *Donner Pass*: As previously discussed, the Donner Pass strategy requires entering other geographical regions where one can act as a new entrant, and attempting to carve a profitable future out of one's incumbent cousins in the industry.
- *Transformer*: Some industry executives we have advised appear very enamored of the idea of finding new and potentially unrelated industries to which they can apply their financial strength and general management expertise. Finding these matches in profitable and defensible industries is harder than it appears and leads to some unusual corporate mission statements. Realistically, not every firm can become a "new media" conglomerate.

There is an important caveat to our work to date. In an effort to separate the structural advantages of new entrants (which even the drastic measures of firing top management staffs, conducting team strategy exercises, and the like cannot change) from the internal difficulties of implementation, we have assumed in our model and our recommended strategies that companies act as quickly and as effectively as they can, subject to the constraints implied by the parameters described above. That is, we assume objective analysis in the process of determining the strategies to pursue and unlimited resolve in implementing these strategies, despite the uncontrolled mirth that these assumptions cause among executives considering change and the organizational change consultants assisting them.

As technology has leapt forward, significant cost and structural advantages have become available to new entrants. Loss of market share by an incumbent cannot be attributed entirely to complacency, to sluggish decision processes and stretched-out reaction times, or to incumbents' outmoded corporate cultures—although these factors exacerbate the problem. Indeed, the very factors that created and defended the strength of dominant players may have become weaknesses that cannot easily be divested. Equally clearly, new management teams and incrementally new strategies alone cannot begin to redress these structural problems; a fundamental rethinking of the burdens of incumbency, along with the potentially sustainable advantages of new entrants, is warranted.

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